



On the time-varying spillover between nonferrous metals prices, geopolitical risks, and global economic policy uncertainty

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Abstract

The present research investigates the dynamic interrelationships between the main nonferrous metal prices (aluminum, copper, lead, nickel, tin, and zinc) and two major risk factors, namely geopolitical risks, and economic policy uncertainty. The analysis employs the TVP-VAR connectedness approach for monthly data from January 1997 to December 2022. The findings may be summarized as follows. First, the analysis confirms the presence of strong spillover effects, which have intensified since the initiation of the financialization of commodity markets during the 2000 s. Second, copper is the most transmitter and receiver of shocks, followed by aluminum and zinc. Moreover, these three metals are net transmitters of shocks for other metals. On the other hand, lead, nickel, and tin are net receivers of shocks from the system. Overall, the study identifies two major clusters of nonferrous metals, aluminum, copper, and zinc, as net transmitters of shocks and lead, nickel, and tin as net receivers of shocks. Third, geopolitical risks and economic policy uncertainty are net receivers of shocks. Moreover, the pairwise connectedness indicates that geopolitical risks are primarily influenced by lead and zinc, while economic policy uncertainty is influenced by all metals, with aluminum being the most prominent. These findings have been shown to be not sensitive to different forecast horizons and lag orders.

Keywords Nonferrous metals prices · Geopolitical risks · Economic policy uncertainty · Connectedness · Time-varying VAR

JEL Classification C58 · G15 · G32 · Q02

1 Introduction

Nonferrous metals play a vital role in economic activity (Gil-Alana and Tripathy 2014; Song et al. 2019). They are an essential input in various industries and significantly impact trade flows, employment, economic growth, and energy transition (Liu et al. 2016; Huang et al. 2020; Lin and Chen 2020; Wang and Feng 2021). The global mining market expanded from \$2022.6 billion in 2022 to \$2145.15 billion in 2023, as Business Research Company (2023) reported. According to Zhang et al. (2022), direct and indirect employment in the Chinese mining industry reached 3.677 million workers in 2019. The mining sector in the United States and Australia creates about 1.2 million direct and indirect jobs (National Mining Association 2016; Australasian Institute for Mining and Metallurgy 2021). Metals are frequently traded commodities, and their exports contribute to foreign exchange earnings. For example, China is the largest exporter of metals, including steel, aluminum, and copper. Steel exports in China accounted for approximately 16% of the global steel export market in 2017 (International Trade Administration 2018). The demand for metals in developed and developing countries has increased rapidly due to fast urbanization and industrialization. For instance, the infrastructure sector in India relies heavily on metals, with steel demand expected to rise from 103 million tons in 2021 to about 430 million tons by 2050 (Sinha and Acharya 2023). Copper, aluminum, zinc, and nickel are among the vital industrial metals crucial in the energy transition process pursued by all countries. Consequently, the demand for these metals is expected to increase significantly in the next decades.

Given the importance of metals for all countries, and with the financialization of commodity markets, it is important for policymakers, investors, and industry participants to understand the dynamics of metal prices and global risk factors influencing them, such as geopolitical tensions and policy uncertainty. Moreover, understanding the spillover effects between the prices of metals, geopolitical risks (GPR) and economic policy uncertainty (EPU) may provide valuable insights into their potential economic repercussions. Metal price fluctuations can influence production costs, investment decisions, and profitability in industries that rely on these resources. Therefore, policymakers and industry participants can better anticipate and manage potential economic challenges by acquiring knowledge of potential spillover effects. Moreover, analyzing this relationship has implications regarding risk management. Indeed, GPR and EPU are significant sources of uncertainty that can affect financial markets. Specifically, political events, such as geopolitical tensions and policy changes, can directly and indirectly affect metals prices. These effects can extend beyond the metal industry and impact broader financial markets, including currencies, stocks, and bonds.

By examining the spillover effects between metals prices, GPR and EPU, investors and risk managers can enhance their risk assessment and develop more effective strategies to mitigate potential losses. In addition, government policies and regulations can substantially impact metals prices. For example, trade policies, tariffs, subsidies, and environmental regulations can affect the supply and demand

dynamics of metals, thereby influencing their prices. Therefore, by studying the spillover effects between metal prices and GPR/EPU, policymakers can also gain insights into the potential consequences of policy decisions. This knowledge can assist policymakers in formulating effective policies that may foster stability and growth in the metals industry. Additionally, international relations can motivate the investigation of such connectedness. Indeed, the global metals market is interconnected due to international trade and supply chains. As a result, metal price changes can affect trade flows and relationships between countries. Henceforth, examining the spillover effects between metal prices and GPR/EPU might provide insights into the interconnectedness and possible risks associated with these relationships. This comprehension can potentially enhance the effectiveness of decision-making in the context of international trade negotiations. In the context of nonferrous metal markets, geopolitical risks and policy uncertainty are significant factors that may impact supply, demand, and price dynamics. Indeed, the global metals industry operates within an environment influenced by geopolitical tensions, policy decisions, and market forces. Therefore, metals price fluctuations may affect sectors, such as construction, manufacturing, transportation, global trade, and economic stability. On the other hand, geopolitical risks, including trade disputes, armed conflicts, and regulatory changes, can potentially disrupt the global metals market. Such risks can impact the production, distribution, and demand for metals, leading to price volatility and market instability. Moreover, economic policy uncertainty, characterized by fluctuations in government regulations, fiscal policies, and monetary interventions, can introduce unpredictability and risk into metal prices, like in all commodity markets.

This study aims to provide novel insights into the interconnections between global risk factors (GPR and EPU) and nonferrous metals (aluminum, copper, lead, nickel, tin, and Zinc) prices between January 1997 and December 2022 using the time-varying parameter vector autoregressive (TVP-VAR) based connectedness approach. By doing so, the study contributes to the existing body of knowledge in many ways.

First, this study adds to the literature by investigating the dynamic interconnection between various nonferrous metals, namely aluminum, copper, lead, tin, nickel, and zinc. Indeed, the dynamic connectedness approach implemented in this study allows identifying which metals transmit shocks to others and which metals receive shocks from others over time. One should note here that recent studies have mainly focused on the interconnections between precious metals. For instance, Tweneboah and Alagidede (2018) analyzed the linkages between gold, platinum and silver using bivariate and multivariate correlations, while Bhatia et al. (2018) conducted a causality-in-quantile analysis to examine the relationship between gold, silver, platinum and palladium. Finally, Sami (2021) explored the long-term linkages between gold and silver in India. Overall, the connections between precious metals, particularly gold and silver, have attracted special attention in past research, owing to the hedging opportunities they offer. On the other hand, limited studies investigated the interconnections between nonferrous metals. For example, Siklos et al. (2020) analyzed the connections between 29 Chinese, Indian, US, and UK futures contracts using connectedness approach. In addition, Gong et al. (2022) examined the

connectedness between six industrial metals and showed that copper, lead and zinc are the main transmitters of shocks in both the spot and futures markets. Furthermore, Mishra and Ghate (2022) investigated the interrelationships between industrial metals from January 2011 to March 2020. The analysis reveals not only a high degree of connectedness between metals but also some mixed findings regarding the direction of spillover. It is worth noting that a growing literature, including Yang et al. (2022) and Bastianin et al. (2023), have instead concentrated on the dynamic linkages between nonferrous metals and other assets/commodities. For example, Urom and Ndubuisi (2023) analyzed the connectedness between industrial metals, precious metals, energy and regional sustainability indices. The analysis suggested that industrial metals are net transmitters of spillovers to other markets. Moreover, Oliyide et al. (2021) explored the connections between major industrial metals (aluminum, iron and copper), precious metals and oil prices. The linkages between oil prices and industrial metals have been also examined by Reboredo and Ugolini (2016), Adekoya and Oliyide (2020), Živkov et al. (2024) and Nasreen et al. (2024), among others. The previous discussion suggests the predominance of empirical studies analyzing the linkages between industrial metals and others metals/commodities, while limited works focused on the interconnections between nonferrous metals. This study contributes to the literature by assessing the interrelationship among several nonferrous metals, offering investors and market participants critical insights into their interactions.

Second, the present research expands upon the existing literature by assessing the potential interactions between various nonferrous metals prices and global risk factors (GPR and EPU) in a common system considered for the first time in literature. Indeed, nonferrous metal prices can be influenced by a variety of factors, including uncertainty, geopolitical events, epidemics, as they are determined in international markets. A survey of the existing literature shows that, unlike precious metals, there is a scarcity of studies examining the impact of global risk factors on the fluctuations of nonferrous metal prices. For instance, Abid et al. (2023) investigated the implications of GPR on five categories of commodities: energy, precious metals, industrial metals, agricultural products and livestock products, using the Markov-Switching model. The findings revealed that GPR does not affect significantly future returns of industrial metals. The primary limitation of this study lies in the use of an aggregate industrial metals index. Considering different metals is important as they may have heterogeneous response to GPR. Wang and Su (2023) examined the interconnections among ten commodity markets, including industrial metals, and five financial markets during the Russian-Ukrainian geopolitical risk period. The TVP-VAR suggests that non-ferrous metals are net transmitters of shocks in both the short- and long-term. Furthermore, Zhao (2023) analyzed the impacts of GPR on various commodity prices, including industrial metals. The results indicated that aluminum, copper, and tin exhibit the most significant response to GPR, followed by lead, nickel, and uranium, with zinc showing the smallest reaction to GPR shocks. The study by Jia et al. (2023) is the only study that analyzed the impacts of both GPR and EPU on the prices of nonferrous metals. However, the study analyzed only the case of China and employed a GARCH-MIDAS model, which does not allow capturing whether each metal is a transmitter or receiver of shocks to other metals and global risk factors. In

addition, Chen et al. (2022) investigated the effects of EPU on the price dynamics of selected nonferrous metals (copper aluminum, lead, zinc, tin, nickel) and precious metals (gold, silver, platinum, palladium) using OLS and quantile regression. The findings suggest that nonferrous metals are significantly affected by EPU. To summarize, one could note that most previous studies employed aggregate measure of industrial metals, and concentrated on analyzing the effects of geopolitical risk or policy uncertainty, but not both (except Jia et al. 2023). This research contributes to the literature by considering simultaneously various nonferrous metal prices, GPR, and EPU in a common system to measure the dynamic connectedness between them.

Third, the study contributes to the literature on commodity price dynamics by exploring the connections between metal prices and global risk factors in a time-varying fashion, allowing to provide more nuanced and fresh insights into the relationships and spillover effects between metal prices, GPR and EPU. The analysis employs the TVP-VAR proposed by Antonakakis and Gabauer (2017) with the connectedness approach developed by Diebold and Yilmaz (2012, 2014). Unlike the traditional VAR models, which do not allow capturing the time-varying dynamics, the TVP-VAR model provides more suitable and informative conclusions on the evolution of interdependence over time. Indeed, the dynamic connectedness approach implemented in this study allows identifying the potential shifts in the role (transmitter/receiver) of each metal throughout the entire period. This is important as the study period is characterized by the occurrence of major events, including the Great Recession, the COVID-19 pandemic, and the Russian-Ukraine conflict. The empirical investigation consists of computing the static and dynamic spillover between nonferrous metals, EPU, and GPR. Then, specific attention is paid to the pairwise interconnectedness between each metal and the two risk factors. Overall, while previous studies explored the connectedness between nonferrous metals, they are limited to specific timeframes and lack comprehensive analysis of the simultaneous impact of GPR and EPU. In contrast, our research extends over a significantly longer period (1997–2022) and employs the TVP-VAR approach to provide a more nuanced understanding of the interconnectedness between nonferrous metals and global risk factors. This broader perspective, along with our methodological approach, allows identifying the role of nonferrous metals as either transmitters or receivers of shocks, a dimension largely absent in prior research.

The rest of the paper is structured as follows. Sections 2 and 3 are reserved for the data and methodology, respectively. In Sect. 4, we present the empirical findings, while Sects. 5 and 6 discuss the empirical findings and concludes the study, respectively.

2 Data and preliminary analysis

The empirical analysis investigates the interplay between several nonferrous metals, GPR, and EPU. To do that, we retrieve monthly returns of nonferrous metals, namely aluminum, copper, lead, nickel, tin, and zinc. The data spans from January 1997 to December 2022, yielding 311 observations and covering many business cycle phases. We use data at monthly instead of daily frequency for many reasons.

First, monthly data has the advantage of being accessible for a long period, which is essential for a better understanding of the connectedness between nonferrous metal price movements and their interaction with GPR and CPU. Second, monthly data allows for the reduction of noise induced by daily data. Indeed, monthly data eliminates short-term market noise by averaging daily fluctuations, thereby facilitating the identification of trends and patterns. Furthermore, the study period covers the major economic, financial, and geopolitical events, including the Asian financial crisis, the Gulf War, the September 11 attacks, the Great Recession, the COVID-19 outbreak, and the Russia-Ukraine conflict. The nonferrous metals price data are obtained from the World Bank Commodity Prices Data, a widely recognized and reliable source for global commodity prices.¹

This paper considers two measures of global risk factors, namely EPU and GPR. The EPU index, developed by Baker et al. (2016), quantifies the extent of uncertainty in economic policy decisions. The index, which is often used as an indicator of market sentiment and investor confidence, is calculated based on several factors, including media coverage of policy-related economic uncertainty, the number of policy-related newspaper articles, and the volatility of stock market returns. Analyzing these factors allows gauging the uncertainty surrounding economic policy decisions. The second global risk measure, GPR, is employed to assess the overall risk and political stability across different countries or regions from a geopolitical viewpoint. It aims to quantify the potential challenges and uncertainties that can arise from political and social factors, including conflicts, policy changes, regulatory developments, and other geopolitical events. The two global risk indices are quantified using numerical values that reflect the level of risk, with higher values indicating higher risks and lower values indicating lower risks. During high EPU/GPR periods, policy decisions become less predictable, leading to cautious investor behavior and reduced business investment. Conversely, a low EPU/GPR index suggests a more stable environment, which could be associated with increased business confidence and investment. For global risk factors, the EPU index was obtained from the Baker et al. (2016) dataset, which measures policy-related economic uncertainty based on newspaper coverage, while the GPR index, which quantifies geopolitical tensions using a range of global political events, was taken from Caldara and Iacoviello (2022).²

Figure 1 displays the patterns of monthly price log returns. We observed the high variability of the different metals returns, especially over the major events: the global financial crisis and the commodity financialization that began in 2008. Specifically, high values are followed by low values and vice-versa, indicating evidence of volatility clustering and asymmetry.

Table 1 provides a summary of descriptive statistics and the correlation matrix. We compute the mean, variance, skewness, and Kurtosis. All series have positive mean values, ranging from 0.136% for aluminum to 0.453% for tin. Regarding the variance, which informs us about the risk level, we find that

¹ <https://www.worldbank.org/en/research/commodity-markets>

² <https://www.policyuncertainty.com/>

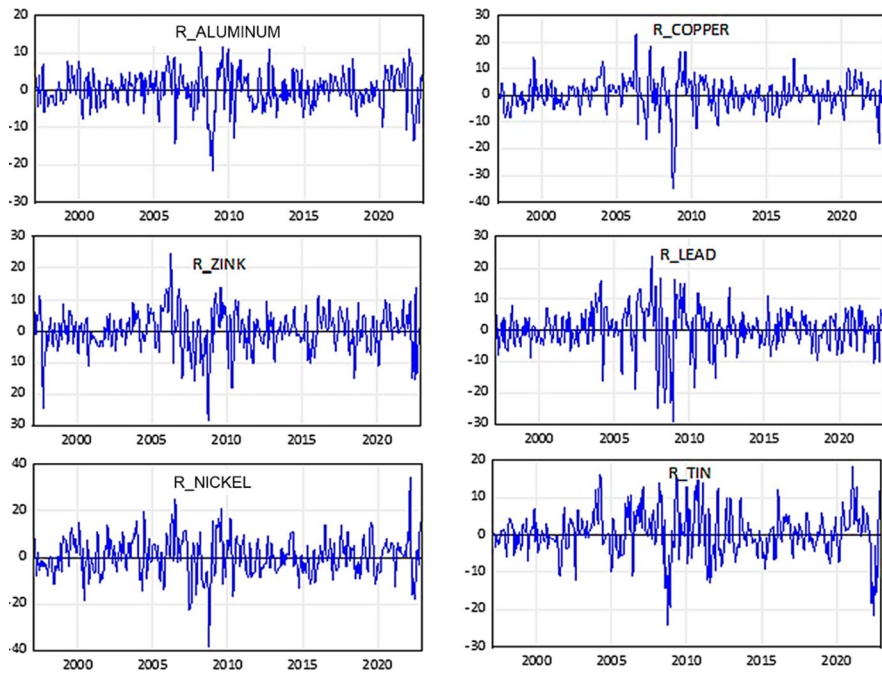


Fig. 1 Monthly returns of nonferrous metal indices

the nickel market is the riskiest, followed by lead market. On the other hand, the aluminum market experienced the lowest risk level during the period. The negative skewness values and significant excess Kurtosis for all metals suggest that the distributions of metal returns are asymmetric and fat-tailed and consequently deviate from the normal distribution. Moreover, the Jarque and Bera statistics strongly reject the normality hypothesis. We also test the presence of unit root test in the metals returns series and the log-difference of the two global risk indices by applying the ERS unit root test proposed by Elliott et al. (1996). The high values of the ERS statistics reject the null hypothesis of a unit root. In addition, the Ljung Box in order 20 shows that all returns series are serially independent for both returns and squared returns, except the GPR squared returns series.

The second part of Table 1 provides the correlation matrix of Kendall tau between the different pairs. We observe the existence of significant correlations between all metals. Aluminum and copper are the most correlated (0.470), while the lowest value is observed between tin and zinc (0.289). Regarding the correlations between the metals returns and GEP/EPU indices, the table shows that metal returns are only correlated with EPU. This suggests that EPU has a more pronounced effect on metal prices than political risk factors.

Table 1 Descriptive statistics, preliminary tests, and correlation matrix

	Aluminum	Copper	Lead	Tin	Nickel	Zinc	GPR	EPU
Mean	0.136	0.397	0.374	0.455	0.453	0.340	0.254	0.379
Variance	23.744***	38.643***	45.418***	38.094***	71.444***	43.523***	487.927***	314.136***
Skewness	-0.545***	-0.665***	-0.611***	-0.283**	-0.093	-0.514***	1.939***	0.672***
Ex. Kurtosis	1.630***	5.313***	2.690***	1.529***	1.593***	1.755***	13.182***	1.768***
JB	49.853***	388.706***	113.105***	34.458***	33.322***	53.606***	2446.398***	63.925***
ERS	-6.704***	-6.516***	-4.152***	-7.257***	-3.648***	-2.970***	-5.255***	-9.709***
Q(20)	43.944***	70.036***	26.699***	65.480***	36.758***	34.056***	27.351***	27.814***
Q2(20)	109.687***	94.557***	178.317***	79.607***	27.744***	33.389***	1.893	12.135
Kendall tau								
Aluminum	1.000***							
Copper	0.470***	1.000***						
Lead	0.322***	0.400***	1.000***					
Tin	0.345***	0.358***	0.301***	1.000***				
Nickel	0.366***	0.383***	0.302***	0.359***	1.000***			
Zinc	0.427***	0.478***	0.434***	0.289***	0.359***	1.000***		
GPR	-0.006	-0.050	-0.063	-0.016	-0.016	-0.074	1.000***	
EPU	-0.123***	-0.105***	-0.097**	-0.066	-0.101***	-0.122***	-0.004	1.000***

*** and ** indicate the statistical significance at the 1 and 5% levels, respectively

3 Empirical methodology

We follow Antonakakis and Gabauer (2017) by applying the time-varying parameter vector autoregression (TVP-VAR) to measure the spillover effects among metals returns and the considered risk factors. Let x_t be a vector of N variables. The TVP-VAR model is structured into the following equations:

$$x_t = \Phi_t x_{t-1} + u_t; u_t | I_{t-1} \sim N(0, S_t) \quad (1)$$

$$\Phi_t = \Phi_{t-1} + v_t; v_t | I_{t-1} \sim N(0, R_t) \quad (2)$$

where I_{t-1} represents the set of information available at $t-1$. Φ_t is a parameters matrix with orders $(N \times Np)$. This matrix is time-varying following a multivariate random walk process described by Eq. (2). The vectors u_t and v_t are $(N \times 1)$ error vectors, having S_t and R_t as $(Np \times Np)$ variance–covariance matrices, respectively, also are time-varying.

To construct the connectedness indices based on Diebold and Yilmaz (2014), the above TVP-VAR model is transformed to its vector moving average (VMA) as follows:

$$x_t = A_t u_t = (A_{1,t} \ A_{2,t} \ \dots \ A_{p,t})' u_t \quad (3)$$

To obtain the connectedness measures based on Diebold and Yilmaz (2014), we use the generalized impulse response functions (GIRF) and the generalized forecast error variance decomposition (GFEVD) developed by Koop et al. (1996) and Pesaran and Shin (1998). To conduct the different spillover indices, Antonakakis and Gabauer (2017) assess the differences between a J -step-ahead forecast where once variable i is shocked and once where it is not shocked as follows:

$$GIRF(J, \delta_{j,t}, I_{t-1}) = E(x_{t+J} | u_{j,t} = \delta_{j,t}, I_{t-1}) - E(x_{t+J} | I_{t-1}) \quad (4)$$

where J be the forecast horizon and $\delta_{j,t}$ be the selection vector equal to 1 on the j th position, and 0 otherwise. Then, the GIRF, which is denoted by $\Psi_{j,t}^g(J)$, can be obtained as:

$$\Psi_{j,t}^g(J) = S_{jj,t}^{-\frac{1}{2}} A_{j,t} S_t u_{j,t} \quad (5)$$

Then, the GFEVD for the horizon J , denoted by $\Pi_{j,t}^g(J)$, can be found as:

$$\Pi_{j,t}^g(J) = \frac{\sum_{t=1}^{J-1} \Psi_{ij,t}^{2,g}}{\sum_{j=1}^N \sum_{t=1}^{J-1} \Psi_{ij,t}^{2,g}} \quad (6)$$

Table 2 Different spillover indices and their interpretations

Index	Formula	Interpretation
Total connectedness index	$H_t^g(J) = \frac{\sum_{j=1}^N \pi_{i,t}^g(J)}{N} \times 100$	Measures the total connectedness between variables in the system
Connectedness to others	$H_{i \rightarrow j,t}^g(J) = \frac{\sum_{j=1}^N \pi_{i,t}^g(J)}{\sum_{j=1}^N \pi_{j,t}^g(J)} \times 100$	Measures the shock that a variable i transmits to variables j .
Connectedness from others	$H_{i \leftarrow j,t}^g(J) = \frac{\sum_{j=1}^N \pi_{i,t}^g(J)}{\sum_{j=1}^N \pi_{j,t}^g(J)} \times 100$	Measures the shock that a variable i receives from variables j .
Net connectedness	$H_{i,t}^g(J) = H_{i \rightarrow j,t}^g(J) - H_{i \leftarrow j,t}^g(J)$	Measures the leading effects of a variable on the network: If $H_{i,t}^g(J) > 0$, the variable i transmits shocks to the network more than it receives If $H_{i,t}^g(J) < 0$, the variable i receives shocks from the network more than it transmits

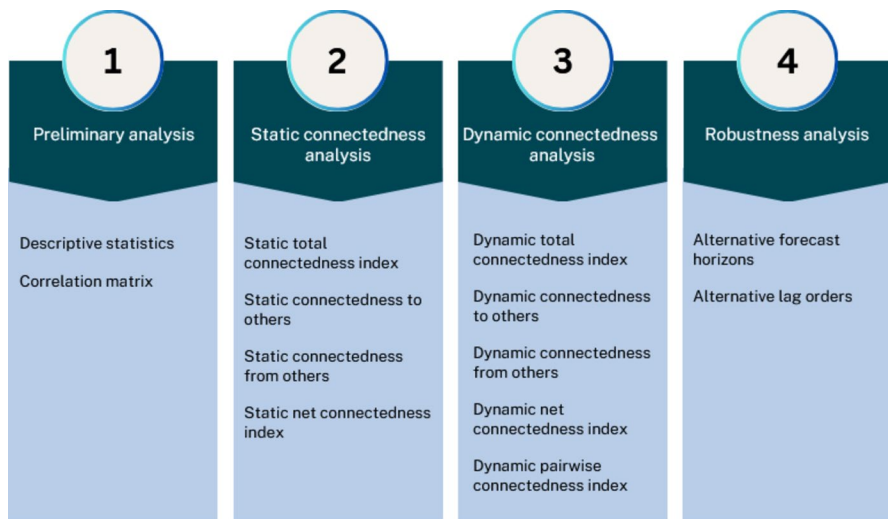


Fig. 2 Empirical methodology

$\Pi_{j,t}^g(J)$ can be explained as the variance share of one variable has on others.³ The GFEVD verifies $\sum_{j=1}^N \Pi_{j,t}^N(J) = 1$ and $\sum_{i,j=1}^N \Pi_{j,t}^N(J) = N$. Using the GFEVD, we can construct the different connectedness indices presented in Table 2.

As shown in Fig. 2, the present study follows an empirical methodology which consists of four stages. We begin with a preliminary check of data properties, including normality, and stationarity, followed by an examination of correlation matrix. Second, we conduct a static connectedness analysis by computing the total, TO, FROM and NET connectedness indices between the considered variables. Third, we re-calculate the aforementioned indices in a dynamic way, which allows obtaining a time-varying total connectedness index, as well as dynamic TO, FROM and NET connectedness indices for each nonferrous metal and risk measure. Furthermore, we also compute the pairwise interconnectedness between each metal and the two considered risk factors. Fourth, we check the robustness of our baseline findings using alternative forecast horizons and lag orders.

³ The shares of variance are normalized in such a way that the sum of each row equals 1. Consequently, all variables collectively account for 100% of the forecast variance of *variable i*, as stated by Antonakakis and Gabauer (2017).

4 Results and discussion

4.1 Static spillover analysis

Table 3 provides the static spillover effects between metal prices, EPU, and GPR indices. The table contains the static pairwise connectedness indices between each pair in the upper part of the table, while the lower part presents the static directional indices “TO”, “Inc. Own”, and “NET”. The last column in the table provides the “FROM” index which indicates the amounts of shocks received by each variable from the system. The diagonal elements of Table 3 indicate the self-contributions, whereas the off-diagonal elements illustrate the interactions among the different variables.

The table suggests that the total connectedness index (TCI) between the considered variables is about 47.19%, suggesting a significant interconnectedness between metal prices and the two global risk factors. When focusing on the role of each metal within the system, the results indicate that copper is the most transmitter (77.99%) and receiver (63.01%) of shocks, followed by aluminum which acts as a transmitter (70.09%) and receiver (61.91%) of shocks, and zinc which is also transmitter (67.32%) and receiver (62.92%) of shocks. These three metals transmit to the system more than they receive, yielding a positive net connectedness index of 14.99% for copper, 8.18% for aluminum and 4.40% for zinc. In contrast, lead, tin and nickel received shocks from the system more than they emit, which means that they are net receivers of shocks from the system. Finally, GPR and EPU are found to be net receivers of shocks, with net connectedness indices of -1.56 and -6.63 , respectively. These findings are in line with Gong et al. (2022) and Umar et al. (2023), who concluded the position of copper and zinc as net transmitters of shocks to other nonferrous metal markets.

Table 3 Static connectedness indices

	Aluminum	Copper	Lead	Tin	Nickel	Zinc	GPR	EPU	From
Aluminum	38.09	17.61	9.53	10.02	10.26	12.99	0.13	1.38	61.91
Copper	15.65	36.99	11.32	9.15	11.02	14.93	0.40	0.54	63.01
Lead	10.82	14.94	42.95	7.30	7.36	15.30	0.56	0.77	57.05
Tin	13.77	12.90	8.83	44.03	10.24	8.95	0.35	0.94	55.97
Nickel	12.88	13.29	7.14	9.60	44.52	11.34	0.26	0.97	55.48
Zinc	13.33	17.15	13.14	7.26	9.85	37.08	1.13	1.07	62.92
GPR	0.48	0.67	1.04	0.21	0.37	1.59	93.39	2.25	6.61
EPU	3.17	1.44	1.56	1.84	2.10	2.22	2.23	85.45	14.55
TO	70.09	77.99	52.56	45.38	51.18	67.32	5.05	7.92	377.51
Inc. Own	108.18	114.99	95.52	89.40	95.70	104.40	98.44	93.37	TCI
NET	8.18	14.99	-4.48	-10.60	-4.30	4.40	-1.56	-6.63	47.19

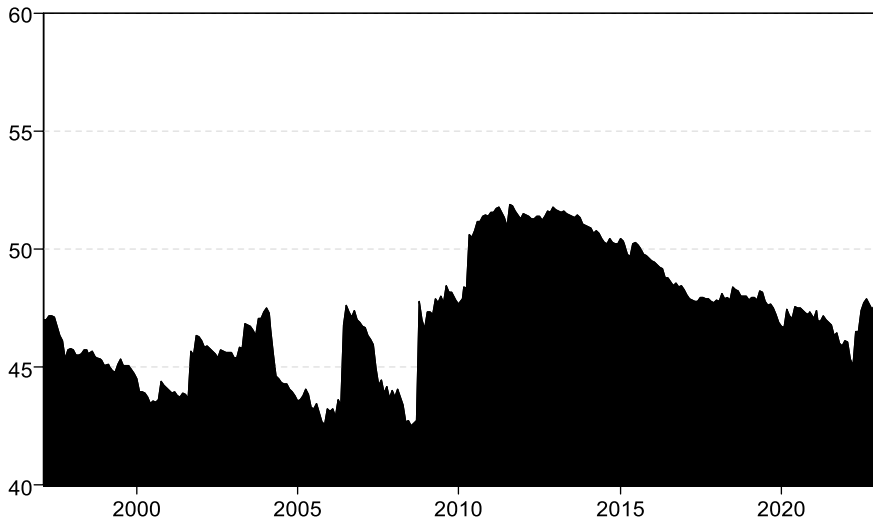


Fig. 3 Dynamic total connectedness between metals returns, EPU, and GPR

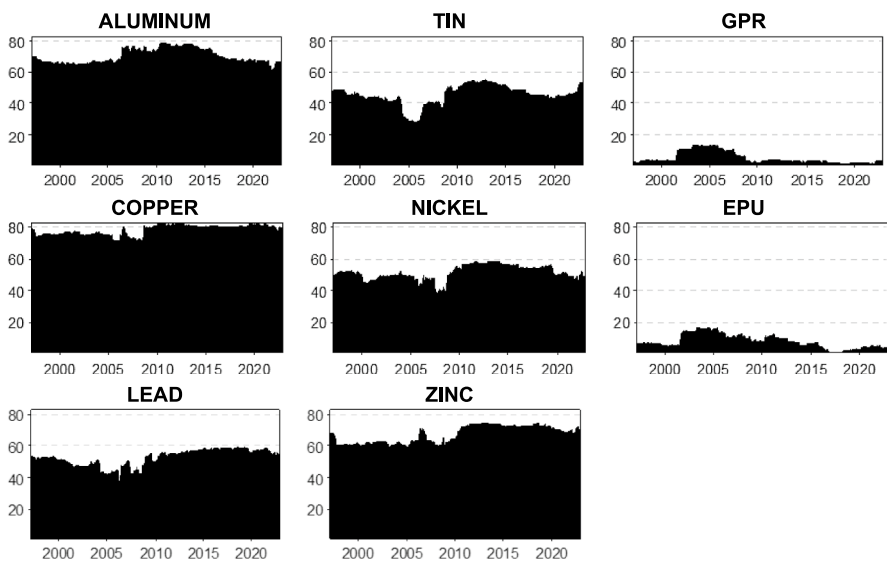


Fig. 4 Dynamic directional connectedness indices from each variable to the system ("TO")

4.2 Dynamic spillover analysis

To provide a clearer picture of the spillover effects between metals prices and the two uncertainty indices, we consider the different spillover indices in a time-varying way. By incorporating time-varying spillovers into the analysis, one can better

understand how spillover effects change in response to different market conditions, shocks, or events. This approach allows for a more nuanced examination of the interconnections and interactions among metals, providing insights into the dynamics of metal markets and their potential spillover effects over time. Figure 3 plots the total connectedness index between the six metal returns, EPU and GPR.

The figure reveals that the spillover effects undergo temporal variations in response to various events that impacted the metals markets. Between 1997 and 2008, TCI exhibited moderate fluctuations, with some periods marked by a decrease in spillover (notably from 1997 to 2001, 2004 to 2005, and 2007), while other periods witnessed an increase in spillover (notably 2002, mid-2003, mid-2006 and 2008) with a TCI ranging between 42 and 47%. However, starting from 2008, there has been a substantial upsurge in shock transmission, which exceeded 50% and reached a maximum in 2012, a period characterized by the rapid financialization of commodity markets. Then, the TCI has decreased progressively starting from 2013. According to Bianchi et al. (2020), such a fall in spillover can be attributed to the de-financialization of the metals market that occurred after the 2013 closure of commodity trading units on Wall Street.

Figures 4 and 5 plot the evolution of the directional spillover indices “TO” and “FROM”. As shown, there is compelling evidence of a time-varying pattern for all metals, CPU, and GPR, which supports the findings of TCI. Figure 4 suggests that the shocks transmitted by the six nonferrous metals are relatively high and exceeded 80% in some cases. For example, copper and aluminum have high shock spillover into the system, which accounted for more than 80% in 2010. Aluminum and zinc also transmitted shocks that exceeded 60% during all the study period. Finally, the shock transmission associated with lead, tin and nickel is less important than other

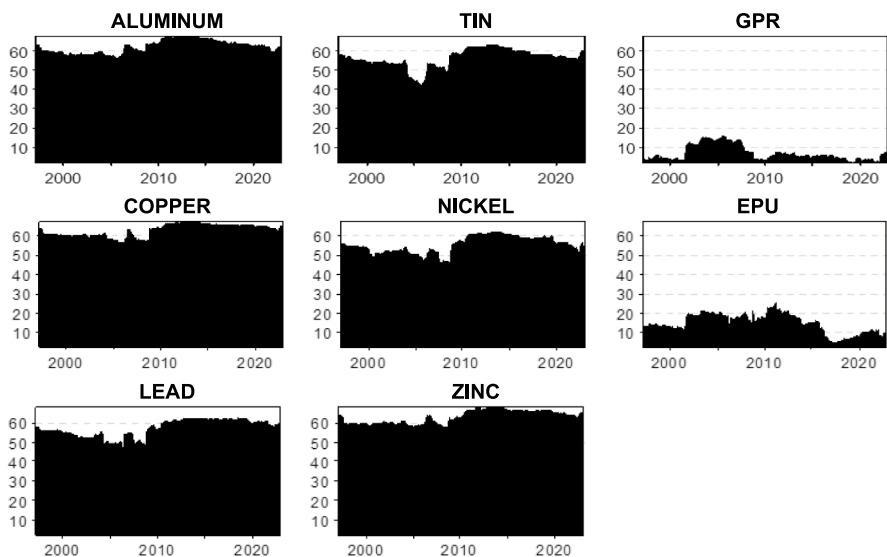


Fig. 5 Dynamic directional connectedness indices from the system to each variable (“FROM”)

metals, while exceeding 50%. Regarding global risk factors, Fig. 4 shows that GPR transmitted shocks particularly between 2002 and 2009, which could be due to the political instability events. These geopolitical events usually lead to a supply disruption, demand fluctuations, risk perception regarding metals prices, and currency and commodity market interactions. Economic policy uncertainty has also transmitted shocks to the system, particularly starting from 2002 and reaching a maximum during the global financial crisis.

At the same time, Fig. 5 suggest that almost all nonferrous metals are the most receivers of shocks from the system. Specifically, the connectedness index surpassed 60% for all metals across different periods, highlighting the significant interconnections among the different nonferrous metals examined. On the other hand, the FROM connectedness indices relative to EPU and GPR has been lower than those of nonferrous metals, remaining around 20%.

Figure 6 depicts the dynamic net spillover index of each variable expressed as the difference between the transmitted and received shocks for each variable within the system. The advantage of this index is that it allows distinguishing between net transmitters and net receivers of shocks over time. In our system, only three metals, aluminum, copper, and zinc, are net transmitters of shocks during the entire period. Our results align with Gong and Xu (2022), who concluded that industrial metals are net transmitters of information to other markets. Furthermore, the findings support those of Ciner et al. (2020), who employed a range of empirical methodologies to emphasize the role of aluminum as a driving force for other nonferrous metals. The prevalence of aluminum, copper, and zinc in information exchange can be attributed to their abundance and extensive utilization in many industries on a global scale. Furthermore, the prices of these metals are influenced mainly by speculative

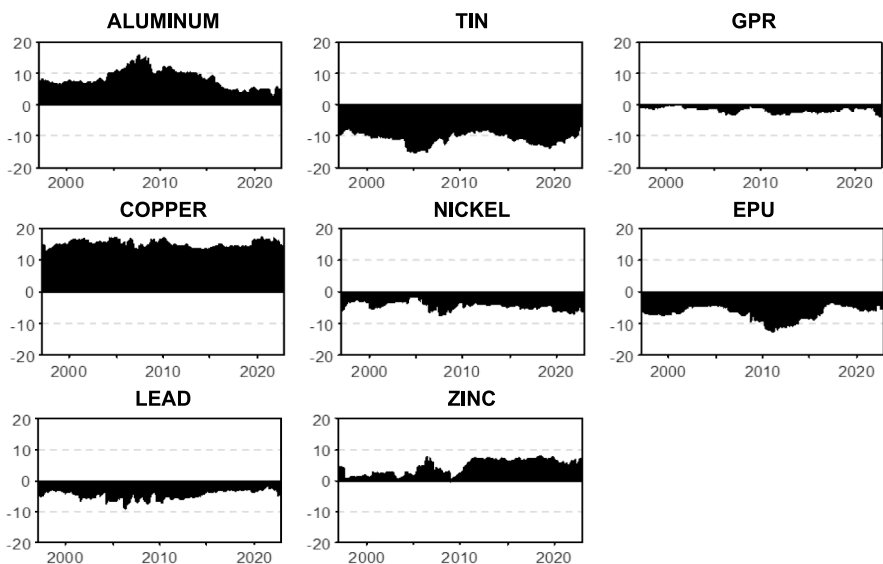


Fig. 6 Dynamic directional “NET” connectedness for each variable

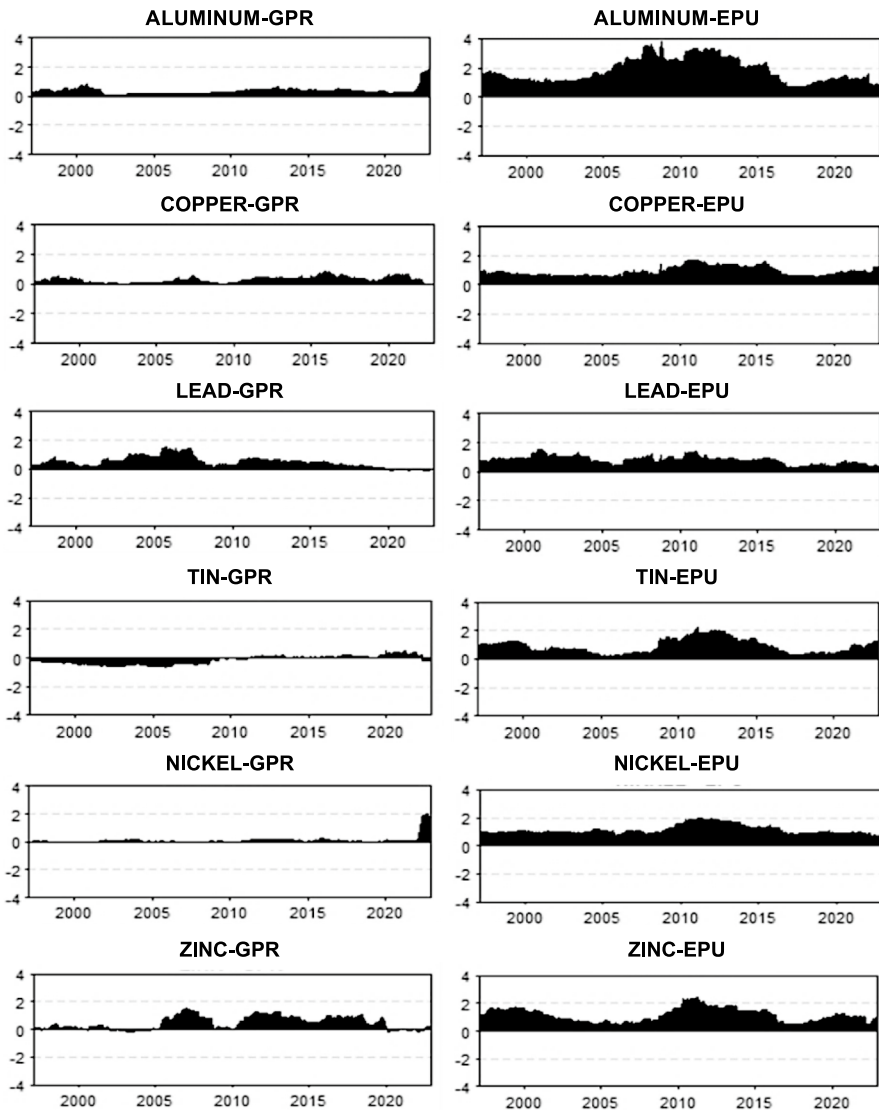


Fig. 7 Pairwise connectedness between metals prices, EPU, and GPR

trading in commodities markets. Investors often turn to these strategic metals to hedge against inflation or diversify their investment portfolios, further impacting their prices.

On the other hand, results show that the two risk factors, EPU and GPR, are net receivers of shocks over the full period. This result can be explained by the economic interconnectedness as metals are used in various industries. Metals market fluctuations can lead to uncertainty and volatility in the broader economy. Moreover, EPU and GPR are often sensitive to market sentiment and perception changes.

When the metals market experiences shocks, investors and market participants may become more cautious and uncertain about the overall economic and geopolitical outlook. They may anticipate potential disruptions in supply chains, shifts in trade dynamics, or geopolitical tensions related to resource access.

4.3 Pairwise connectedness between nonferrous metals markets and GPR/EPU

This section discusses the pairwise connectedness between nonferrous metal prices and the two risk factors. The findings illustrated in Fig. 7 highlight several important conclusions. First, the interconnectedness between nonferrous metals markets and GPR/EPU varies over time and depends on the considered metal and risk measures. Therefore, there is a substantial heterogeneity in the interconnectedness between the nonferrous metals and global risk factors. This heterogeneity represents an argument for considering different metals and risk measures. Second, the outcomes reveal that all nonferrous metals prices—risk factors pairwise connectedness indices are positive over the entire period. This confirms that nonferrous metals markets transmit more shocks to EPU and GPR than they receive. These outcomes are in accordance with those in Fig. 6, which suggest that some nonferrous metal markets are net transmitters of shocks, while EPU and GPR are receivers of shocks over the entire period. However, Fig. 7 adds some new insights into the contribution of each metal to EPU and GPR. Third, we find that the amount of shocks transmitted from the different metal markets to the EPU is more significant than GPR. In other words, economic policy uncertainty has been demonstrated to be more sensitive to all nonferrous metals than geopolitical risk over the entire period. Fourth, lead and zinc have been identified as the most significant metals transmitting shocks to GPR. Regarding EPU, the figure confirms the role of aluminum as a primary transmitter of shocks to uncertainty. The amount of shocks transferred by aluminum to the EPU index is the highest compared to all other metals and risk measures. In addition, the connectedness reached its maximum during the global financial crisis. Uncertainty has also been affected by shocks in zinc, tin, and nickel prices. Although lead and copper are net transmitters of shocks to EPU, they have the lowest spillover over the entire period.

4.4 Robustness analysis

In this section, we conduct a robustness analysis in a series of steps to verify the previous findings. We first re-estimate the TVP-VAR model and calculate the different connectedness indices by considering different forecast horizons, namely 20, 30, and 40 months. Then, we employ different lag orders of the estimated model, i.e., 2, 3 and 4. Tables 4 and 5 provide the average connectedness indices (TCI, TO, FROM, and NET) by changing the forecast horizon and lag order, respectively. In addition, Figs. 8 and 9 depict the dynamic total connectedness index for different forecast horizons and lag orders, respectively.

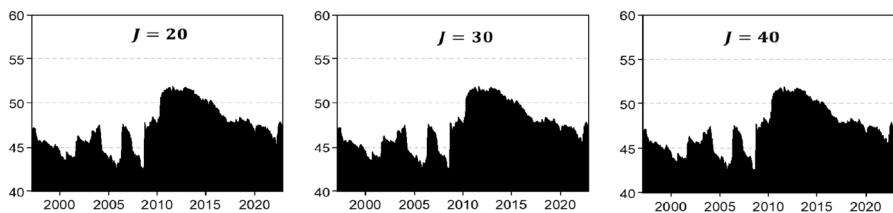
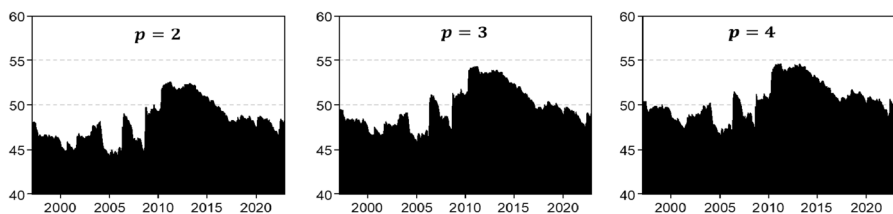
According to Table 4, the connectedness indices for various lag orders indicate that the copper, aluminum and zinc are the main transmitters of shocks to the

Table 4 Connectedness indices using different forecast horizons (J)

	Aluminum	Copper	Lead	Tin	Nickel	Zinc	GPR	EPU
$J=20$								
To	TCl=47.19 70.09	77.99	52.56	45.38	51.18	67.32	5.05	7.92
From	61.91	63.01	57.05	55.97	55.48	62.92	6.61	14.55
Net	8.18	14.98	-4.49	-10.59	-4.30	4.40	-1.56	-6.63
$J=30$								
To	TCl=47.19 70.09	77.99	52.56	45.38	51.18	67.32	5.05	7.92
From	61.91	63.01	57.05	55.97	55.48	62.92	6.61	14.55
Net	8.18	14.98	-4.49	-10.59	-4.30	4.40	-1.56	-6.63
$J=40$								
To	TCl=47.19 70.09	77.99	52.56	45.38	51.18	67.32	5.05	7.92
From	61.91	63.01	57.05	55.97	55.48	62.92	6.61	14.55
Net	8.18	14.98	-4.49	-10.59	-4.30	4.40	-1.56	-6.63

Table 5 Connectedness indices using different VAR lag orders (p)

	Aluminum	Copper	Lead	Tin	Nickel	Zinc	GPR	EPU
$p=2$	TCI=48.14							
To	67.19	77	53.05	49.65	52.41	68.3	5.24	12.26
From	63.88	63.71	56.91	57.27	56.06	63.18	7.08	17
Net	3.31	13.29	-3.86	-7.62	-3.65	5.12	-1.84	-4.74
$p=3$	TCI=49.72							
To	68.03	77.97	57.4	48.93	54.57	70.96	6.31	13.55
From	64.56	64.69	58.16	59.1	57.59	63.9	10.64	19.1
Net	3.47	13.28	-0.76	-10.17	-3.02	7.06	-4.33	-5.55
$p=4$	TCI=50.51							
To	67.13	77.24	56.83	51.91	56.37	71.53	8.12	14.97
From	64.93	64.46	58.1	59.88	57.62	63.06	14.87	21.18
Net	2.2	12.78	-1.27	-7.97	-1.25	8.47	-6.75	-6.21

**Fig. 8** Total connectedness indices between metals prices, EPU, and GPR using different forecast horizons (J)**Fig. 9** Total connectedness indices between metals prices, EPU, and GPR using different VAR lag orders (p)

system. Specifically, copper is the most transmitter of shocks (14.98%), followed by aluminum (8.18%) and zinc (4.40%). On the other hand, lead, tin and nickel are net receivers of shocks. In addition, GPR and EPU are found to be net receivers of shocks, with net connectedness indices equal to -1.56 and -6.63. These findings are consistent with those presented in Table 3 and highlight that results are not sensitive to the selected forecast horizon. Table 5 also illustrates that the findings the main conclusions still hold when considering different lag orders. The net connectedness indices suggest that copper is the most transmitter of shocks, followed by aluminum

and zinc, while tin, lead and nickel are net receivers. Furthermore, although having different net indices for different lags, GPR and EPU both have negative net indices. This emphasizes that GPR and EPU receive more shocks from others than they transmit to the system.

Figure 8 reports the time-varying total connectedness indices for $J=20$, $J=30$ and $J=40$. As can be observed, the indices have the same pattern over time and are similar to our baseline results reported in Fig. 3. In addition, Fig. 9 indicates a similar trend regarding the total connectedness indices for lag orders ranging between 2 and 4. Overall, the results reported in this section indicate no significant differences with baseline findings when using alternative forecast horizons and lag orders, thereby confirming the robustness of results.

5 Discussion

The main objective of this research is to analyze the static and dynamic connectedness between nonferrous metals, GPR and EPU. The empirical investigation yielded some interesting findings, which can be outlined as follows. First, the dynamic connectedness analysis revealed a substantial upsurge in shock transmission starting from 2008. This corresponds chronologically to the financialization of commodity markets and the onset of the global financial crisis (Henderson et al. 2015; Bianchi et al. 2020; Zaremba et al. 2021; Ghazani et al. 2023). The increased connectedness among metal markets and the two global risk factors that began with the financialization of commodity markets and the global financial crisis can be attributed to many factors. First, the influx of financial investors into commodity markets enhanced markets integration, making various commodities more interconnected. As a result, shocks or changes in one commodity market, such as metals, can transmit more easily to other markets (Zaremba et al. 2021). The increased integration may, therefore, amplify spillover effects between metals prices and other factors, including EPU and GPR. Second, following the financialization of commodity markets, including the metals markets, financial investors have often engaged in short-term trading and speculative activities, leading to higher price volatility. Then, increased liquidity can magnify price movements and make them more sensitive to external factors such as EPU and GPR. Consequently, changes in these risk factors can have a more pronounced impact on metals prices. Third, another factor explaining this increase in interconnections is herding behavior and sentiment-driven trading. Indeed, financialization can contribute to herding behavior, where investors follow prevailing trends or sentiments rather than fundamental supply and demand factors. This behavior can amplify price movements and increase the transmission of spillover effects. In the context of EPU and GPR, shifts in market sentiment driven by economic or geopolitical events can lead to heightened trading activity and more significant spillovers between metals prices and these factors. Fourth, the financialization of commodity markets is often accompanied by the development of derivative products, such as commodity futures and options, which allow investors to speculate or hedge their exposure to commodities. The trading and pricing dynamics of these derivatives may influence the underlying commodity markets and facilitate spillover

effects. Moreover, the pricing of commodity derivatives may incorporate factors like EPU and GPR, further linking them to metals prices. Finally, it is important to note that the magnitude of connectedness began to fall starting from 2013, but it remained higher than in 2008. This can be attributed to the de-financialization of metals markets, among others, from 2014 to 2017, after the 2013 closure of commodity trading units on Wall Street (Bianchi et al. 2020).

Second, both static and dynamic connectedness analyses suggested copper as the biggest transmitter of shocks to the system, followed by aluminum and zinc. In addition, these three nonferrous metals are net transmitters of shocks for all other metals. These findings partially corroborate those of Umar et al. (2023), who employed the TVP-VAR model to examine the connections between the same set of nonferrous metals (aluminum, copper, lead, nickel, tin, and zinc). The authors demonstrated that copper, zinc, and tin are net transmitters of shocks to other nonferrous metals, while lead and aluminum are net receivers. Our findings are also in line with those of Gong et al. (2022), who analyzed the connectedness between the six industrial metals mentioned above and concluded that copper, zinc and lead are the main transmitters of shocks in both the spot and futures markets. Therefore, a common conclusion between our results and the above-mentioned studies is that copper and zinc are net transmitters of shocks to other nonferrous metals. The dominance of copper, zinc and aluminum in shock transmission can be explained by many factors. First, these metals account for the largest share of global industrial metals consumption and are commonly traded metals used in multiple industries (World Bank 2021). The high trading volumes and extensive market participation make them more sensitive to external factors and news events, leading to a higher likelihood of spillover effects. In contrast, lead, nickel, and tin may have comparatively lower liquidity and trading volumes, which can limit the magnitude and speed of spillover transmission. Second, these strategic metals are widely used in various industries. Therefore, changes in demand or economic conditions within these industries can significantly impact the prices and market dynamics of these metals. Although also utilized in different applications, lead, nickel, and tin may have lower demand, which can reduce the extent of spillover effects. Third, financial and speculative activities, including investment strategies and trading behavior, can influence the metals market. Aluminum, copper, and zinc have historically attracted more attention from investors and speculators, resulting in increased price volatility and the potential for spillover effects. According to Al-Yahyaee et al. (2020), aluminum and zinc have the highest trading volume among metals on the London Metal Exchange. Moreover, there is an intensified demand for copper, aluminum, and zinc since they are considered among the industrial metals crucial in the rapid deployment of clean energy technologies as part of the energy transition process (International Energy Agency 2022). Furthermore, copper is the predominant nonferrous metal heavily utilized for renewable energy investments. It is crucial in producing electric vehicles, wind turbines, solar photovoltaic systems, and nuclear technologies. In contrast, lead, nickel, and tin may have lower speculative interest, leading to comparatively less pronounced spillover transmission.

Finally, the results suggest that GPR is a transmitter of shocks to nonferrous metals, while not very important, between 2002 and 2009. This could be attributable to

the political instability and conflicts in some regions where metals are sourced (the Iraq invasion in 2003, the Madrid bombing in 2004, and the London subway bombing in 2005; Johannesburg violence acts, among others). These geopolitical events usually lead to supply disruption, demand fluctuations, risk perception regarding metals prices, and currency and commodity market interactions. Political tensions and policy uncertainty, along with financialization, have led to heightened spillovers between the prices of nonferrous metals and global risk factors. Our results corroborate previous studies, such as Naeem et al. (2021) and Mokni et al. (2021). Moreover, the findings confirm those of Ciner et al. (2020), who revealed that the co-movements between the London Metal Exchange nonferrous metal market have increased after the global financial crisis.

6 Concluding remarks and implications

The present study investigated the complex interconnection between six nonferrous metal prices, namely aluminum, copper, lead, nickel, tin, and zinc, and the two most significant global risk factors, geopolitical risk, and economic policy uncertainty, from January 1997 to December 2022. Using the time-varying VAR-based connectedness approach of Diebold and Yilmaz (2012, 2014), the findings underscore a dynamic spillover effect between the metals markets and EPU and GPR, which was amplified following the financialization of commodities markets and the onset of the global financial crisis. Moreover, the findings confirm that aluminum, copper, and zinc are net transmitters of shocks over time. On the other hand, the other metals and the two global risk factors are net receivers of shocks.

The conclusions of the present study may serve as the basis for providing recommendations to policymakers, market participants, and stakeholders. First, the results highlight the role of some nonferrous metals as net transmitters of shocks. Therefore, there is a need to promote economic diversification in countries heavily reliant on nonferrous metal exports, thereby mitigating the adverse impacts of metal price volatility. The promotion of alternative economic sectors may reduce the vulnerability of countries to fluctuations occurring in international nonferrous metal markets. In addition, diversification can help mitigate the impact of shocks in metal markets on the overall economy. Second, the findings reveal that aluminum, copper and zinc are the most influential nonferrous metals, serving as net transmitters of shocks to other metals, EPU, and GPR. Therefore, designing and implementing strategies aiming at stabilizing the price of these strategic industrial metals is of utmost importance, as this stabilization is transmitted to other nonferrous metals and global risk factors. This can be achieved, for instance, by minimizing reliance on a specific country/region as a source of metals, and fostering international cooperation to establish transparent regulations aimed at stabilizing the aforementioned nonferrous metals markets. Third, the findings highlighted the existence of substantial connections between the nonferrous metal markets, GPR, and EPU. Consequently, it is essential for policymakers and market participants to account for this interdependence between all three components (metal markets, GPR and EPU) when intervening in any one of them. This could provide more effective strategies

to mitigate risks in international markets. Fourth, investors and companies involved with nonferrous metals or facing price volatility may implement risk management practices and adopt hedging strategies to reduce the exposure to market fluctuations. Achieving this objective may involve the use of suitable financial instruments, the development of commodity futures markets and the promotion of these risk management tools among market participants.

Although the present study provides fresh insights into the dynamic connectedness between nonferrous metal prices, GPR, EPU, there are some limitations that should be further investigated. First, while the used TVP-VAR model has many advantages, alternative methodologies, such the quantile VAR, could be used to provide connectedness measures that account for different markets conditions. Second, future research may conduct a region-specific analysis to identify potential differences in the interdependence between geopolitical risks, economic policy uncertainty and nonferrous metal markets. Finally, other measures of risks may be considered, including climate policy uncertainty, trade policy uncertainty and energy policy uncertainty. Considering other uncertainty measures may offer valuable findings on the interdependence between nonferrous metal markets and global risk factors.

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Declarations

Conflict of interest The authors have no conflicts of interest to declare.

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